

Consistency of Linear Systems of Equations

High-Level Overview

When dealing with a system of linear equations, one of the most fundamental questions in engineering mathematics is: **Does a solution exist?**

Depending on the answer, a system can be:

- **Consistent:** Possesses **one or more** solutions.
- **Inconsistent:** Possesses **no** solutions at all.

To determine consistency systematically, we convert a system of linear equations into matrix form ($AX = B$) and analyze the ranks of the coefficient matrix and the augmented matrix using **Rouche's Theorem**.

1. Matrix Representation of Linear Systems

Consider a general system of m linear equations in n unknowns ($x_1, x_2, \dots, x_n$):

$$\begin{aligned}a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\&\vdots \\a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m\end{aligned}$$

We can break this system down into three main matrices:

1. **Coefficient Matrix (A):** An $m \times n$ matrix containing the coefficients of the variables.

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}_{m \times n}$$

2. **Variable Matrix (X):** An $n \times 1$ column vector of the unknowns.

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n \times 1}$$

3. **Constant Matrix (B):** An $m \times 1$ column vector of the right-hand side values.

$$B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}_{m \times 1}$$

Thus, the entire system can be compactly expressed as a matrix equation:

$$AX = B$$

The Augmented Matrix $[A \mid B]$

By appending the constant column vector B to the right of matrix A , we form the **augmented matrix**:

$$[A \mid B] = \left[\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{array} \right]$$

2. Homogeneous vs. Non-Homogeneous Systems

- **Homogeneous System:** If all constants on the right-hand side are zeros ($b_1 = b_2 = \dots = b_m = 0$), the system is written as $AX = 0$.
 - *Note:* A homogeneous system always has at least one solution called the **trivial solution** ($x_1 = 0, x_2 = 0, \dots, x_n = 0$).
- **Non-Homogeneous System:** If at least one constant b_i is non-zero ($AX = B$).

3. Rouché's Theorem for Consistency

Rouché's Theorem states that a system of linear equations is consistent **if and only if** the coefficient matrix A and the augmented matrix $[A \mid B]$ have the exact same rank. Otherwise, the system is inconsistent.

Summary of Conditions (Let n = number of unknowns, and let r be the rank):

Condition	Nature of System	Number of Solutions
$\text{rank}(A) = \text{rank}(A \mid B) = n$	Consistent	Unique Solution
$\text{rank}(A) \neq \text{rank}(A \mid B)$	Inconsistent	No Solution
$\text{rank}(A) = \text{rank}(A \mid B) < n$	Consistent	Infinite Number of Solutions

Deep Dive into the Cases:

1. **Unique Solution:** Occurs when the rank of both matrices equals the number of variables.
2. **No Solution:** Occurs when row reductions lead to a contradiction (e.g., $0 = 5$), making $\text{rank}(A) \neq \text{rank}(A \mid B)$.
3. **Infinite Solutions:** Occurs when the rank is less than the number of variables ($r < n$). In this case, we assign arbitrary parameters to $n - r$ of the unknowns and express the remaining r unknowns in terms of those parameters.

Quick Exam Revision Summary

- **Matrix Form:** Always translate $AX = B$ and build the augmented matrix $[A \mid B]$.
- **Rouché's Theorem Rule:** Compute the ranks using row operations (Echelon form).
 - $\text{rank}(A) = \text{rank}(A \mid B) \implies$ Consistent (Solution exists).
 - $\text{rank}(A) \neq \text{rank}(A \mid B) \implies$ Inconsistent (No solution).
- **Solution Count:**
 - If Rank = Variables $\rightarrow$ Unique.
 - If Rank < Variables $\rightarrow$ Infinite.